Reading the Moon from a Lunar Thin Section

The Apollo 11 Sample 10020

Figure 1: Photo of Apollo 11 sample 10020 taken in the F-201 during initial processing. This sample is 6cm across, covered with micrometeorite craters. There are 16 thin sections of sample 10020. Photo credits: Lunar Sample Compendium

This lunar rock 10020 was brought back by the Apollo 11 astronauts from Mare Tranquilitatis, one of the Moon's dark basaltic plains. When you look up at the Moon, you can clearly distinguish two contrasting regions: the dark maria and the bright highlands. The maria appear as dark patches, which are actually vast plains of solidified basaltic lava; the highlands are the lighter regions that surround the maria, representing the Moon's older crust.

Petrographic Observations of Sample 10020

Lunar rocks are made up of minerals and glasses. A mineral is a naturally occurring, solid substance with a specific chemical composition and an ordered crystal structure. Glasses are solids that may have compositions similar to minerals, but they lack the ordered internal arrangement of atoms. By slicing a rock into a thin section, typically about 30 micrometers thick, and placing it under a polarizing microscope (that can control how light passes through a rock thin section), scientists can observe how minerals interact with light. This interaction reveals a wealth of information about mineral identity, crystal structure, and the history of the rock itself.

In transmitted light microscopy, thin sections are examined using two main viewing modes: plane-polarized light (PPL) and crossed-polarized light (XPL). PPL uses one polarizing filter below the stage as light vibrates in a single direction as it passes through the sample thin section. Minerals are examined for basic properties such as color, relief, transparency (vs opacity), grain shape and boundaries, and cleavage in PPL. XPL uses two polarizing filters: one below the sample and the other above it, oriented at 90° (crossed). When the microscope is switched to XPL, a different set of features becomes visible. Many minerals display vivid interference colors caused by birefringence, the splitting of light into two rays traveling at different speeds through the crystal. As the stage is rotated, these colors change, and minerals periodically go dark in a phenomenon known as extinction. These behaviors are key to identifying minerals and understanding their crystallographic orientation. Twinning patterns can also be seen (e.g., stripes in plagioclase) under XPL.

Figure 2: Leica DM750 P polarizing microscope

A sequence of PPL and XPL images of sample 10020 at 5° intervals was acquired from Virtual Microscope. These images were subsequently aligned to ensure a consistent field of view during stage rotation. Anisotropic minerals exhibit periodic optical behavior in XPL, with extinction and interference patterns repeating every 90°, resulting in four repetitions over a full 360° rotation. As a result, the changes in optical properties of individual grains can be directly observed during stage rotation.

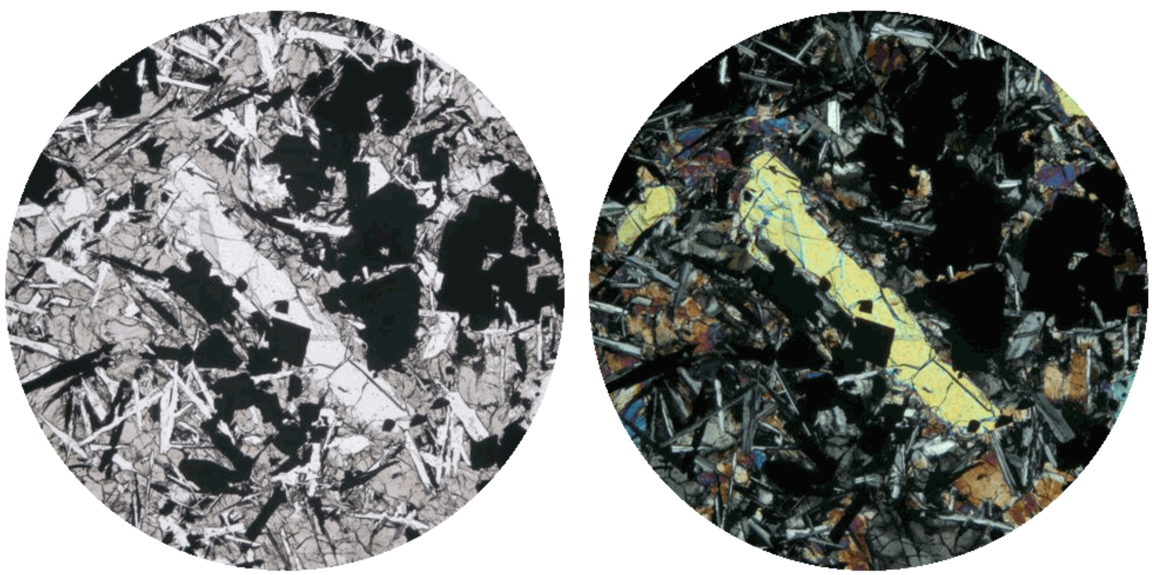

Figure 4: Sequence of PPL/XPL images of sample 10020 thin section Photo Credits: Virtual Microscope

Key observations from the rotation sequence using both PPL and XPL together of the same field of view (2mm in diameter) are as follows:

1. Central bright grains

In PPL:

- Appearance: Appearing bright / white; no pleochroism
- Relief: High relief, Grain boundaries appear sharp and stand out.
- Shape (habit): Grains look blocky / irregular.
- Texture: Fractured, with irregular cracks
- Other features: They show irregular internal boundaries consistent with adjoining grains.
- They are surrounded by finer groundmass. They locally contain small dark grains and small clear / cloudy grains.

In XPL:

- Interference colors: Relatively high-order inference colors (vivid colors that shift as the sample is rotated)
- Extinction: straight extinction (parallel to crystallographic axes)

Interpretation:
The bright centerpiece is best interpreted as a clustered olivine phenocryst aggregate, consisting of multiple subgrains with similar optical orientations. Olivine grains locally contain inclusions of (1) opaque minerals, likely trapped during crystal growth; and (2) likely melt inclusions.

2. Pale brown grains

In PPL:

- Appearance: Pale brown (groundmass), subdued appearance
- Relief: Moderate to high relief
- Shape: Blocky
- Cleavage: straight, parallel lines within the grains
- Other features: They are distributed throughout the groundmass. Some occur along the margins of the centerpiece.

In XPL:

- Interference colors: Moderate interference colors
- Extinction: Uniform extinction

Interpretation:
The pale brown grains scattered throughout are likely pyroxene.

3. Lath-shaped crystals

In PPL:

- Appearance: Bright
- Relief: Low relief (harder to see boundaries). They look less crisp.

- Shape: Elongated laths
- Other features: They form a network of laths throughout the groundmass. They appear less distinct than higher-relief minerals. They are intergrown with pale brown grains.

Under XPL:
- Interference colors: Low-order interference colors (gray, white) when not in extinction.
- Polysynthetic twinning: They appear as alternating light and dark stripes (most diagnostic feature).
- Extinction: They go extinct in segments. During rotation, one set of stripes goes dark (extinction), and the adjacent stripes are still bright, and then they switch; which creates a striped flashing pattern.

Interpretation:
The lath-shaped crystals are plagioclase.

4. Completely black grains in all frames

- Appearance: Complete black in both PPL and XPL at every rotation angle
- Other features: They are distributed throughout and also occur as inclusions within the central grains. They occupy / fill the spaces between earlier-formed minerals. They are relatively large, abundant, and range from acicular forms to blocky forms.

Interpretation:
These are opaques (likely ilmenite or Fe-Ti oxides in lunar samples). The relatively coarse and abundant oxide minerals (likely ilmenite) occur between earlier-formed silicate grains, suggesting crystallization from a Fe–Ti-rich melt during the later stages of cooling. The inclusions are likely Cr-spinel (chromite).

5. Small irregular bright grains

- Appearing bright / white in PPL.
- Irregular shapes
- Relatively high interference colors in XPL
- Not dominant in this specific field

Interpretation:
These are possibly minor olivine.

6. Melt inclusions

- Appearing tiny, rounded or irregular inside a host mineral
- Clear or cloudy in PPL and black / mixed in XPL

Interpretation:

These are likely melt inclusions, tiny trapped droplets of ancient melt inside centerpiece olivine.

7. Glasses

- Appearing white in PPL but always black in XPL

8. Overall texture

This thin section of sample 10020 shows olivine as relatively large grains, plagioclase as lath-shaped crystals forming a framework, pyroxene occurring between plagioclase and along olivine margins, and oxides present between silicate grains. These observed relationships are consistent with a porphyritic basalt texture in which early-formed olivine is enclosed within a finer-grained groundmass. The arrangement of plagioclase laths and pyroxene suggests an intergranular to locally subophitic texture. The occurrence of pyroxene along olivine margins indicates later crystallization from the remaining melt, while the presence of relatively coarse oxide minerals reflects crystallization from a Fe-Ti-rich melt during the later stages of cooling. Together, these features record a sequential crystallization history during progressive cooling of a basaltic melt.

Petrographic observations provide a first-order identification of minerals based on optical properties; however, definitive mineralogy is established through chemical and crystallographic analyses (Nesse, 2012; Deer et al., 2013). Based on optical characteristics and textural relationships, the minerals in this thin section of sample 10020 are interpreted as olivine, pyroxene, plagioclase, and Fe-Ti oxides. These identifications are consistent with previous chemical and microanalytical studies, which confirm this mineral assemblage (Meyer, 2011). Each mineral carries information about the conditions under which it formed, and their collective arrangement reveals the history of the rock. In sample 10020, the textural relationships between olivine, pyroxene, plagioclase, and oxide minerals allow reconstruction of a crystallization sequence and cooling history. These interpretations extend beyond mineral identification, providing insight into the processes of magma evolution and, ultimately, the thermal evolution of the Moon.

The Moon may seem still and cool, but it was once thermally active, and alive with molten rock flowing across its surface and shaping its geology.

Segmentation of Opaques

You may want to give it a try to segment opaque phases in the PPL and XPL images, allowing the texture of the sample to be visualized more clearly:
https://huggingface.co/spaces/pingwang2026/segmentation-of-opaques-in-petrographic-images